



Aerodynamic Loads Identification and Modeling of UCAV Configurations with Control Surfaces Using Prescribed CFD Maneuvers

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This work is focused on developing aerodynamic models that rapidly and accurately predict stability derivatives of several test cases over a wide range of flight conditions. The test cases are unmanned air vehicles that share the same planform but differ in their span-wise profile shapes. The planform was developed under the NATO AVT-251 Task Group, referred to as the MULDICON (or MULti-Disciplinary CONfiguration). Aerodynamic loads models are only developed for normal force and pitch moment coefficients, however, the developed approach can easily be extended to include lateral aerodynamic coefficients as well. Quasi-steady models of this work are power series expansions of traditional linear aerodynamic models to capture non-linear effects. Additionally, the models can estimate static and dynamic stability derivatives and the control powers. Models attempt to capture global aerodynamics, as they cover changes in angle of attack (α) from -5° to 15° , a Mach range from 0.2 to 0.8, and inboard and outboard control surface deflections between $\pm 20^\circ$, while also independently modeling $\dot{\alpha}$ and q . Schroeder signals are used for the training maneuvers, which excite the parameters orthogonally and simultaneously. The accuracy of the developed models are compared to static data generated by CFD at test points across the flight-envelope, as well as against dynamic off-design maneuvers performed in CFD. All simulations are completed using the *Kestrel* CFD software. Once the models are created, they can generate all aerodynamic data needed for aircraft stability and control analysis on the order of a few seconds.

Nomenclature

C_m	pitching moment coefficient
$C_{m\alpha}$	pitching moment curve slope, 1/rad
C_{mq}	pitch moment due to q , 1/rad
$C_{m\dot{\alpha}}$	pitch moment due $\dot{\alpha}$, 1/rad
C_N	normal force coefficient
$C_{N\alpha}$	normal force curve slope, 1/rad
C_{Nq}	normal force due to q , 1/rad
$C_{N\dot{\alpha}}$	normal force due to $\dot{\alpha}$, 1/rad
c	characteristic length
k	reduced frequency, $\frac{\omega c}{2V}$
M	Mach number
q	normalized pitch rate
t^*	normalized time step, $\frac{tV}{c}$
V	free-stream velocity, m/s

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Greek

α	effective angle of attack, deg
$\dot{\alpha}$	normalized time rate of change of α
β_i	fit coefficient
δ_i	control surface deflection, deg
θ	pitch angle, deg
ω	angular frequency, rad/s

I. Introduction

There is a growing attempt to close the gap between aircraft Stability & Control (S&C) analysis and computational fluid dynamics (CFD). The availability of high-fidelity aerodynamic predictions from CFD during early stages of aircraft design has the potential to help in accurate design of control laws, thereby avoiding costly modifications in downstream stages. A number of challenges arise when using CFD in aircraft design. One significant challenge is aircraft details, such as junction/joint geometries, are not available during the conceptual design phase. The application of Reynolds-Averaged Navier-Stokes (RANS) equations on these rough geometries can be misleading as separated flows can occur at these junctions. Ghoreyshi and Cummings [1] showed that this can affect overall coefficient values. However, the general design trends can still be predicted by applying CFD to these geometries. Another challenge is how accurately CFD can predict relevant parameters for aircraft S&C. Nelson et al. [2] showed that although CFD cannot replace experimental tests for generation of S&C data sets, it offers the potential for high fidelity S&C analyses and thus a greater understanding of aircraft dynamics much earlier in the design life cycle, saving time and cost in the long term. A final issue is related to the computational cost of generating a CFD data set for S&C study. This often requires several thousands of simulations. Many efforts to reduce this computational cost and accelerate the generation of such a data set have been demonstrated including the work of authors [3–5]. Specifically, Allen and Ghoreyshi [5] presented several prescribed maneuvers coupled with aerodynamic models that accurately and rapidly predict the aerodynamic stability derivatives of a generic missile configuration over a wide range of air speeds using CFD simulations. The current work is a follow-up to Ref. [5] to develop aerodynamic models for several unmanned combat aerial vehicle (UCAV) test cases with moving control surfaces. Some of this work originally appeared in Ref. 6.

These UCAVs were designed within NATO AVT-251 Task Group. The objective of the NATO AVT-251 Task Group [7] is to computationally design a new MULti-Disciplinary CONFIGuration (Mulidcon), from previous NATO Task Group experiences to meet a set of desired mission requirements. A number of designs are evaluated, and each design shares a similar planform but has unique cross-sections. Designs considered in this work are “Design 1”, “Design 2”, and “Design 3”. Design 1 was created by Raj Nangia and the Defence Science and Technology Laboratory (DSTL) in the United Kingdom. Design 2 has the airfoil cross sections of the AVT-161 configuration, called the SACCON, and is often referred to as the baseline. The objective of this design was to computationally design a flyable vehicle based on the baseline planform to meet a desired mission. For example, the design should have a lift coefficient of 1.0 during takeoff and landing ($M = 0.2$) and reach $C_L = 0.5$ at high altitude and high Mach number ($M = 0.8$). These requirements cannot be met by the AVT-161 SACCON configuration or the Muldicon baseline design. Design 3 is very similar to Design 2 but has a modified wing center section to allow integration of a single turbo-fan engine with a fan diameter of about one meter. The aerodynamic performances of the baseline design and Design 3 are compared in Ref. 8. The authors concluded that Design3 wing indicates an improvement in sustained turn performance and is a considerable improvement over the baseline wing. This work is the first effort to apply system identification methods using prescribed motions to aerodynamic modeling of a UCAV concept with moving control surfaces.

A Schroeder-type forced motion is used to cover angle of attack, Mach number, and control surface deflections of inboard (IB) and outboard (OB) flaps of these UCAVs. In more detail, the input space covers changes in angle of attack (α) from -5° to 15° , a Mach range from 0.2 to 0.8, and inboard and outboard control surface deflections between $\pm 20^\circ$, while also independently modeling $\dot{\alpha}$ and pitch rate, q . The accuracy of the developed models are compared to static data generated by CFD at test points across the flight envelope, as well as against dynamic off-design maneuvers performed in CFD. The paper is organized as follows: The system identification methods are described; The CFD solver and setup are detailed; The prescribed training maneuver is defined; Finally, the system identification results and summary are presented.

II. Methodology

A. Aircraft System Identification

Determining the functional dependencies between the inputs and outputs of the aircraft system is called aircraft system identification (SID). Once these functional dependencies are established, other analyses can be completed, including S&C analyses. System identification is a wide field in terms of the methods used to determine and model the functional relationships. Time dependent, quasi-steady, and frequency domain methods are all used to perform aircraft SID. This work uses a quasi-steady method, also referred to as a derivative-based method, where functional relationships are developed as linear polynomial equations in terms of the aircraft flight parameters and rates. No direct dependence on time is retained.

A quasi-steady model is limited in its applicability where the aircraft might experience fully unsteady flight characteristics, like dynamic stall or flutter. A useful measure for characterizing the unsteadiness of fluid flow is the reduced frequency, $k = \omega c / 2V$, where ω is the oscillation frequency, c is the characteristic length, and V is the freestream velocity. Generally, reduced frequencies below $k = 0.01$ are considered quasi-steady, and while results from Greenwell [9] demonstrate this, his results also demonstrate that dynamic derivatives can show large variations with reduced frequency. This demonstrates that the quasi-steady model will begin to become inaccurate as unsteady flow effects begin to dominate the aerodynamics. Nonetheless, a quasi-steady model is very useful for estimating the main trends in an aircraft's aerodynamics.

The quasi-steady SID modeling in this work follows a specific process. First a training maneuver (a computational 'test-flight') is completed in CFD. The training maneuver will vary any flight parameters of interest, often simultaneously, to generate training data. Once the maneuver is complete, models can be constructed using any desired technique by comparing the variations of the inputs with the calculated integrated loads (the aerodynamic coefficients). In this work, a dynamic training maneuver is completed because the effects of q and $\dot{\alpha}$ are of interest. The longitudinal integrated loads on the aircraft are studied in this work, C_N and C_m . Once the models are constructed, they are used to simulate an 'off-design' maneuver. The off-design maneuver is not used to train the models but acts as a comparative data set. The training and off-design maneuvers should be different to show that the generated models are extensible to different types of maneuvers.

While parameter selection techniques can be applied to the model selection process, such as the multivariate orthogonal functions demonstrated by Klein and Morelli [10], none were applied in this work. The reason is that control over the specific terms which are included in the model must be maintained. Since the models cover the control surface influence over the Mach and angle of attack range of interest, in addition to static and dynamic stability derivatives, very specific terms are needed to ensure that estimates could be extracted for all these desired parameters. The resulting models are validated against static CFD results, dynamic CFD results, and off-design CFD maneuvers.

The models start from a traditional linear aerodynamic model, shown here for C_N :

$$C_N = C_{N_0} + C_{N\alpha}\Delta\alpha + C_{N\dot{\alpha}}\dot{\alpha} + C_{Nq}q + C_{N\delta_i}\Delta\delta_i \quad (1)$$

Next, the tradition model stability terms can be extended by multivariate power series to capture changes of these terms over the Mach and angle of attack range of interest, demonstrated for $C_{N\alpha}$:

$$C_{N\alpha}(M, \alpha) = \beta_0 + \beta_1(M + \alpha) + \beta_2(M + \alpha)^2 + \dots + \beta_n(M + \alpha)^n \quad (2)$$

As this equation is further expanded, all scalar factors from the binomial expansions are absorbed into the β coefficients, and any like-terms are combined to give:

$$C_{N\alpha}(M, \alpha) = \beta_0 + \beta_1 M + \beta_2 \alpha + \beta_3 M^2 + \beta_4 M \alpha + \beta_5 \alpha^2 + \beta_6 M^3 + \beta_7 M^2 \alpha + \beta_8 M \alpha^2 + \beta_9 \alpha^3 \dots \quad (3)$$

Every term from Eq. 1 can be expanded similarly to capture behaviors over any variables of interest. Any terms that show up twice can be combined to have a common β fit coefficient. The order of the power series expansion can be arbitrarily selected to capture expected curvatures of each stability coefficient. However, a higher order of expansion will yield more terms, increasing the opportunity for over-fitting. The final models should be compared against off-design maneuvers to ensure that they are not over-fit.

For this work, a first-order expansion for $C_{N\dot{\alpha}}$, C_{Nq} , and $C_{N\delta_i}$ terms was used. Third and fourth-order order Mach and α expansions were used for $C_{N\alpha}$ and C_{N_0} respectively. The final model used for all coefficients is given by:

$$\begin{aligned}
C_a = & \beta_0 + \beta_1\alpha + \beta_2\alpha^2 + \beta_3\alpha^3 + \beta_4\alpha^4 + \beta_5M + \beta_6M^2 + \beta_7M^3 \\
& + \beta_8M^4 + \beta_9M\alpha + \beta_{10}M\alpha^2 + \beta_{11}M\alpha^3 + \beta_{12}M^2\alpha + \beta_{13}M^2\alpha^2 + \beta_{14}M^2\alpha^3 \\
& + \beta_{15}M^3\alpha + \beta_{16}M^3\alpha^2 + \beta_{17}q + \beta_{18}\dot{\alpha} + \beta_{19}Mq + \beta_{20}M\dot{\alpha} + \beta_{21}\alpha q + \beta_{22}\alpha\dot{\alpha} \\
& + \beta_{23}\delta_{OB} + \beta_{24}\delta_{IB} + \beta_{25}M\delta_{OB} + \beta_{26}M\delta_{IB} + \beta_{27}\alpha\delta_{OB} + \beta_{28}\alpha\delta_{IB} \quad (4)
\end{aligned}$$

where, in this work $a = N, m$.

This same model was used for all MULDICON planforms. In this form, the fit coefficients are solved by linear regression. All modeling is implemented using MATLAB.

B. CFD Solver

The *Kestrel* CFD solver, KCFD, was used for all simulations in this study. *Kestrel* was developed under the DoD High Performance Computing Modernization Program's (HPCMP) Computational Research and Engineering Acquisition Tools and Environments (CREATE™) program. *Kestrel* is part of the CREATE Air Vehicles (CREATE-AV) software initiative. The purpose of the CREATE program is to develop multi-disciplinary physics-based computational engineering tools that will increase the efficiency of the acquisition process. The *Kestrel* software specializes in fixed wing, full vehicle aerodynamic analysis.

Kestrel was developed with a modular architecture, such that new capabilities can be added to the code as they are developed with relative ease. This allows the user to expose and use capabilities as needed, thereby making the software applicable to many areas of aircraft aerodynamic analysis. The KCFD solver, used in this work, is a finite-volume cell-centered solver which can run with mixed element type unstructured meshes. The Method of Lines is used to separate the temporal and spatial integration. This allows a second-order accurate Godunov scheme to be employed in space, as well as a second-order subiterative point-implicit scheme in time.

Arbitrary prescribed body maneuvers (which are rigid-body transformations of the mesh relative to the reference flow) can be specified in *Kestrel* through a user-generated text file. *Kestrel* accepts three formats for this file. For this work, the 'Euler Angle' file type was used. The motion file is formatted by listing the Euler angles and any mesh translations at discrete and increasing times. For this work, new angles and translations are specified at the simulation time step. Using this format for motion specification allows for the aircraft pitch, and subsequently q , to be explicitly specified, while allowing for an independent angle of attack to be achieved by mesh translations relative to the reference flow.

A body hierarchy is used in *Kestrel*, which lets the user define parent and child bodies. This hierarchical structure applies any transformations of the parent body automatically to the child bodies. In addition, any forces acting on the the child bodies are automatically added to the parent body. Overset control surfaces can be added as a special type of child body to the parent aircraft. Arbitrary control surface deflections can be specified using another motion file, similar to the arbitrary prescribed body motion. The arbitrary control surface motion file allows the hinge line and the angle of deflection to be specified at discrete and increasing times. Details about the *Kestrel* software can be found in the *Kestrel* user's guide [11].

C. CFD Settings

Many static CFD cases are completed to create a set of validation data to compare to the SID modeling. Static cases are completed at zero angle of attack for three Mach numbers, 0.2, 0.4, 0.8. At Mach 0.4, three additional angles of attack are studied: 5, 10, and 15 degrees. At each of these conditions, five control surface deflections are studied: baseline, IB = -20, IB = +20, OB = -20 and OB = +20. Also at Mach 0.4, IB and OB deflections of ± 10 degrees are added. Finally, at Mach 0.2 and 0.8, simulations of $\alpha = 5$ degrees are completed with no control surface deflections.

All static CFD cases are run for 3500 iterations, with data averaged over the final 1500 iterations. A time step of $t^* = 0.01$ is used for all static simulations in this work. 600 start-up iterations are used to initially converge the flow. Temporal damping of 0.005, and three subiterations are used.

In addition to the static CFD cases, some dynamic CFD cases are completed to estimate the dynamic stability derivatives. For these cases, a single oscillation of both pitching and a plunging is completed at Mach 0.2, 0.4, and 0.8. Pitching motions are used to estimate the q effects, and plunging oscillations are used to estimate the $\dot{\alpha}$ effects. Oscillatory motions are commonly used to estimate the dynamic stability derivatives using CFD, which follows a similar

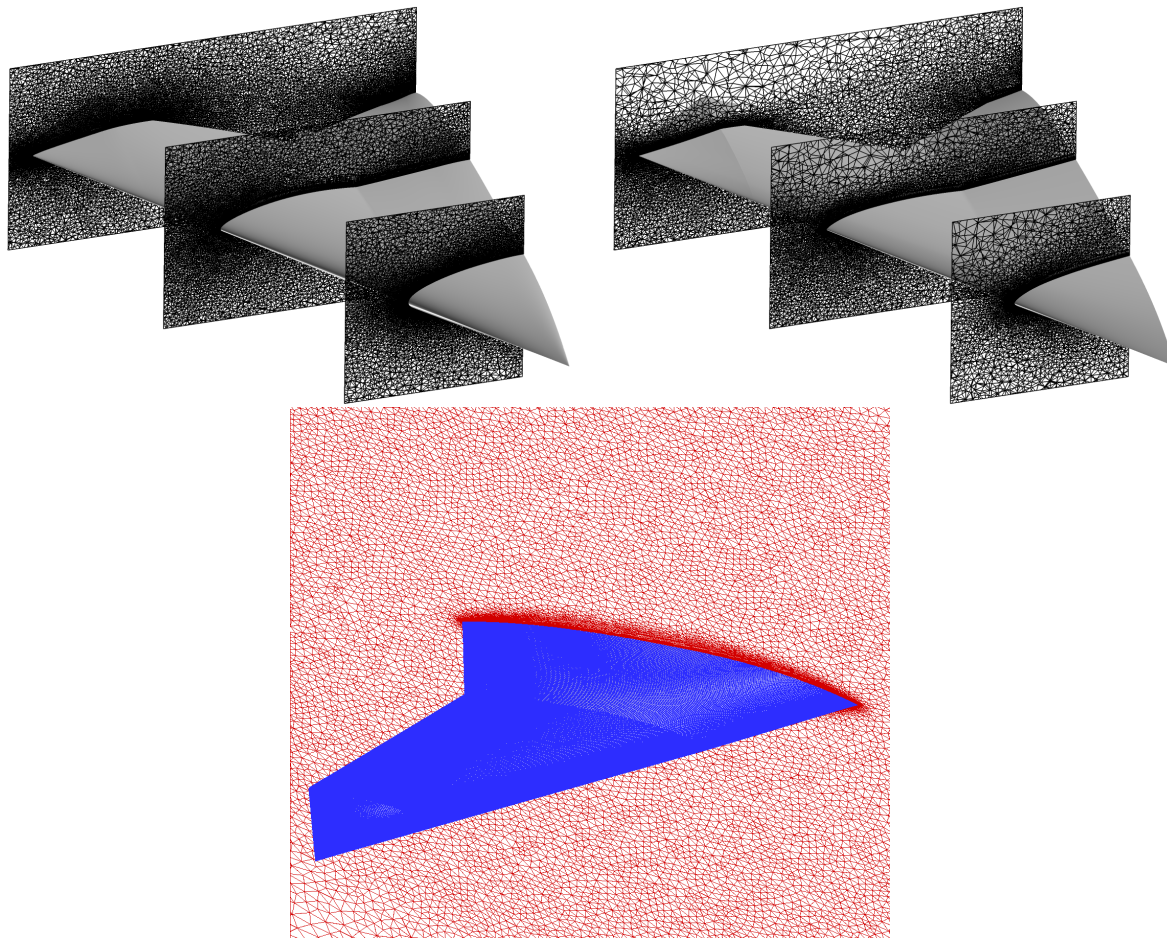


Fig. 1 First row: X-plane cuts of the meshes used for Design 1 (left) and Design 2 (right). Second row shows surface and symmetry mesh for Design 3 wing.

practice for wind tunnel tests. The single oscillations are executed at $k = 0.075$, so simulation lengths vary from 23000 to 83000 iterations, depending on the freestream Mach condition, with time step of $t^* = 0.01$. Higher temporal damping is used on the dynamic simulations, at 0.15. Also, 5 subiterations are used to increase the temporal accuracy for moving mesh simulations.

For the SID training maneuvers, the same CFD settings are used compared to the dynamic CFD single oscillation cases. However, the SID maneuvers are 40 seconds in length, thus requiring 229172 iterations. All CFD cases used the SARC+DDES turbulence model, with which Nelson *et al.* [2] showed good accuracy for the similar SACCON configuration in *Kestrel*.

The meshes used for Muldicon designs 1, 2, and 3 are shown in Figure 1. Overset meshes were used for the control surfaces.

D. Training Maneuver Generation

The training maneuvers used in this work are constructed from Schroeder signals [12]. Schroeder showed that harmonic, multi-sine signals could be optimized by peak factor to achieve a low-amplitude, mixed frequency signal. Morelli [13] extends the optimized Schroeder signal to N variables by combining every N^{th} harmonic frequency. Each resultant set of harmonic frequencies is separately optimized following Schroeder's process. These signals are orthogonal and have low correlation, which allows them to be applied to separate aircraft parameters and varied simultaneously.

For this work, all Muldicon designs complete the same 40 second training maneuver. The maneuver inputs are shown in Figure 2. Four Schroeder signals are generated to vary α , θ , IB , and OB deflection. Simultaneous to these

inputs, the reference Mach condition is steadily changed from 0.1 to 0.9. Extending the Mach range to just beyond the range of interest (Mach 0.2 to 0.8) can increase the accuracy of the final models. This is executed by translating the mesh into the oncoming flow, increasing the relative Mach number of the aircraft. Morelli demonstrates the technique of changing reference condition ‘behind’ a training maneuver in Ref [14], with good success.

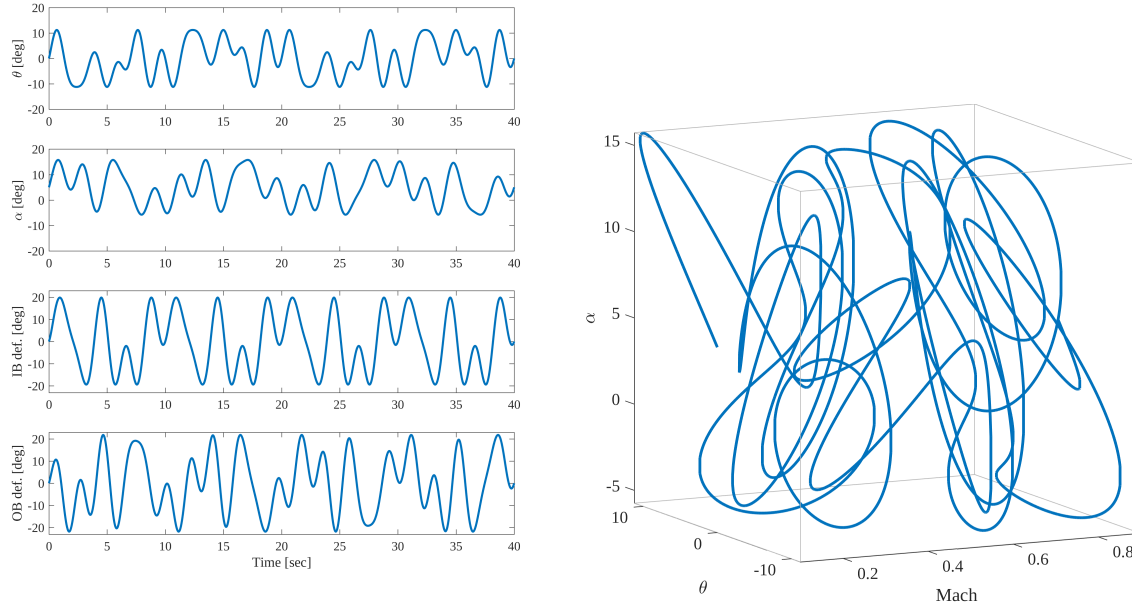


Fig. 2 The Schroeder signals during the 40 second training maneuver (left), and the Mach- α - θ regressor coverage (right).

The Schroeder signals are generated using the SIDPAC [15] `mkmswp.m` function. These signals are constructed from the harmonic frequencies ranging from the lowest frequency limited by simulation length, up to $k = 0.075$. The angle of attack is set to vary by 20° , from -5° to 15° , and θ are set to vary around 0° by 20° as well. The control surface deflections are set to vary by 40° , between $\pm 20^\circ$. This results in a maximum control surface deflection rate of 57.8° per second. The maximum pitch rate is 32.3° per second, and the maximum angle of attack rate was similar.

The training maneuver varies α independent of θ as this allows for the independent modeling of $\dot{\alpha}$ and q effects. This is achieved by varying α through ‘plunging’ maneuvers relative to the reference flow. The magnitude of the plunging changes as the reference Mach changes. Care is taken to ensure the magnitude of the velocity past the aircraft increased linearly and smoothly even with the additional translations needed to vary α and Mach.

III. Results

A. Baseline CFD Results

Static CFD runs are completed for Muldicon designs 1 and 2. These results act as a baseline which the SID models will be compared against. Only a SID training maneuver is completed for Muldicon Design 3. This section only describes Design 1 and Design 2 results. SID predictions of Design 3 are shown in the Appendix.

The notable static results from the Muldicon designs 1 and 2 deal with the control surface deflections. One of the goals of the Muldicon planform shape is to increase the control surface effectiveness compared to the previous planform design called the SACCON. The SACCON experienced poor control surface effectiveness due to significant span-wise flow over the SACCON control surfaces at high angles of attack [16]. The reduction in the trailing edge sweep angle lessens the tendency of the flow to move laterally at high- α conditions and also turns the control surface hinge lines more normal to the flow. Both of these factor allow for more effective control surfaces for the Muldicon planforms.

Four angles of attack were completed at Mach 0.4 for designs 1 and 2, and at each condition, five control surface

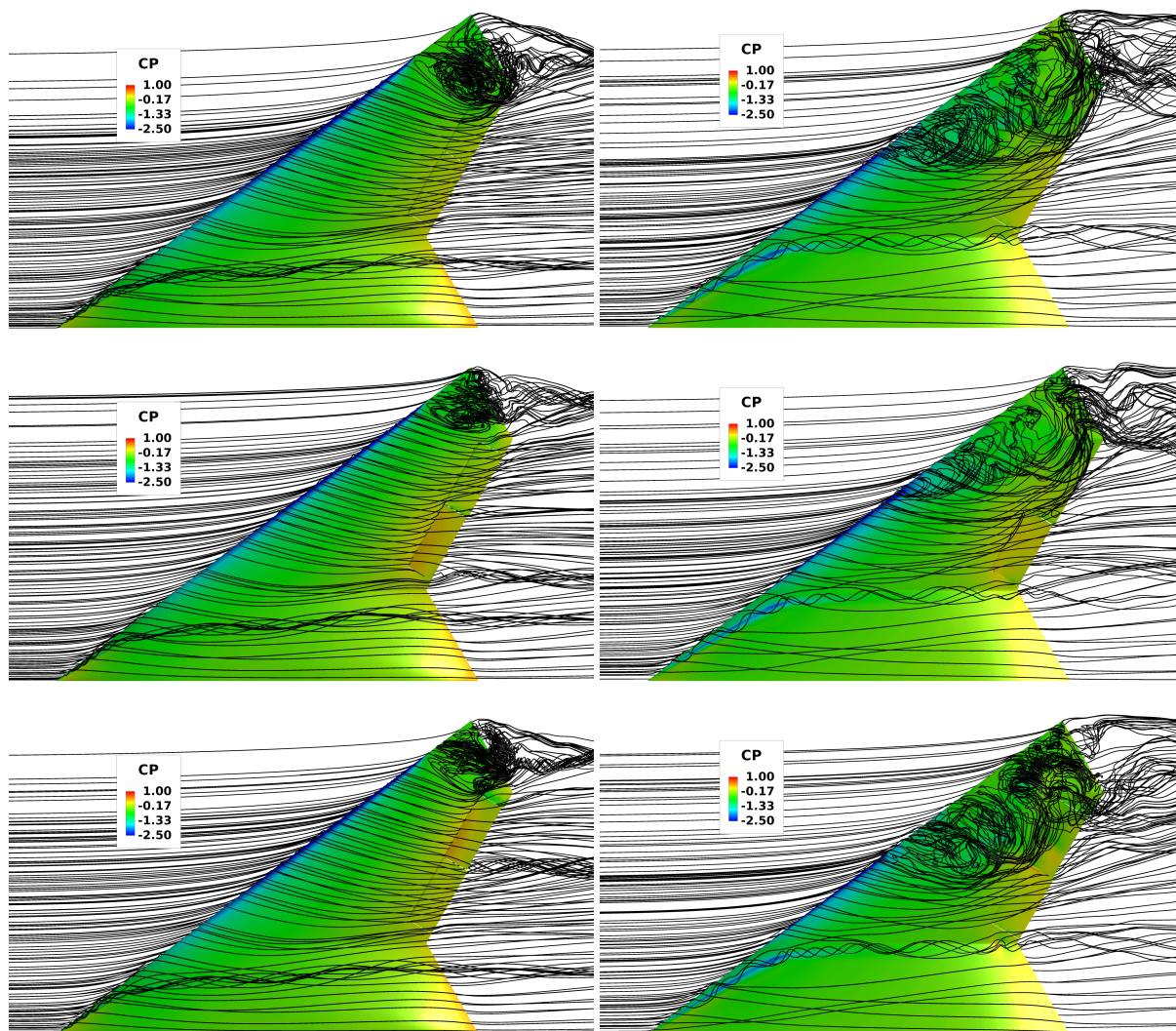


Fig. 3 Design 1 (left column) and Design 2 (right column) at Mach 0.4 with no deflection (top row), IB = -20° (middle), and OB = -20° (bottom).

settings were simulated. Figure 3 compares the flow for both designs at $\alpha = 15^\circ$, with no deflections, $OB = -20^\circ$, and $IB = -20^\circ$. This figure shows via streamlines one of the main differences between design 1 and design 2, namely, a significant reduction in separated flow at the wingtip at high- α . The separated flow of Design 2 covers the outboard control surface significantly at this condition, reducing the control surface effectiveness. This behavior is most pronounced for negative control surface deflections.

The other static results are shown with the SID modeling results, labeled in the figures as ‘Static CFD.’ The dynamic oscillation results from the single pitching and plunging oscillations are also shown with the SID modeling results. These are labeled in figures as ‘Dynamic CFD.’ It should be noted that these dynamic CFD oscillation cases were run at relatively high reduced frequencies and only for a single oscillation. Normally, these would be completed for 2 or 3 oscillations at reduced frequencies at or below 0.01. As such, the values returned for the dynamic CFD derivative are comparative in nature and don’t necessarily represent ‘truth’ data.

B. SID Model Results

The 40 second Schroeder type training maneuvers for designs 1 and 2 were completed on DoD HPCMP machines, Garnet and Excalibur. On Garnet for Muldicon design 1, the total simulation took 1.34 million CPU hours. The SID model fits for C_N and C_m during the training maneuver are shown in Figure 4. These show good matching to the training maneuver data, with normalized root mean squared error below 3.5%. The quasi-steady models do not capture the transient flow unsteadiness which is seen especially for C_m CFD data in Figure 4. This is one issue when using quasi-steady models for aircraft SID, where unsteadiness in the integrated loads is common. However, this can be dealt with by fitting another quasi-steady model to a measure of the unsteadiness of the aerodynamic coefficients to help quantify the variation in the models. No effort was made in this work to do this, however, Morton and McDaniel demonstrate the technique in Ref [17].

Once constructed, the SID models are used to predict the aerodynamic behavior over the flight envelope tested by the static simulations. This includes the ability to predict the control surface effects, compared between the two configurations. Comparisons between the static control surface deflections at varying α for C_N are shown in Figure 5 and for C_m are shown in Figure 6.

Figure 5 shows that the C_N behavior is quite linear for IB and OB control surface deflections up to $\pm 20^\circ$ for both Muldicon designs. The SID models do a good job of predicting the coefficient values at these conditions, as well as control surface effectiveness which is apparent by the slope of the lines.

Figure 6 shows what was described earlier in the static results section. Specifically, the loss of control surface effectiveness for negative OB deflections at high angles of attack. In this case, the SID models do capture the non-linearity of the control surface effectiveness with increasing α for Design 2, and also predict some loss of effectiveness for Design 1 which is not apparent in the static CFD data. Overall, the SID models show quite close matching to the static CFD results, with a few exceptions. Additionally, the trends of the control surface effectiveness are captured reasonably well by the models.

The SID models also can be differentiated to recover estimates of the static and dynamic stability derivatives. The results comparing the SID models with the static stability derivatives (calculated by finite-differencing of static CFD simulation results) are shown in Figure 7. The overall trends of the static stability derivatives are captured very well by the SID modeling. More static points would be needed to determine quantitatively how accurate the SID models are at predicting the static stability. Additionally it should be noted that the training maneuver used for the SID models creates something similar to a ‘central-difference’ estimate of the static stability at $\alpha = 0$ since it fits data from α of -5 up to 15 degrees. It would be of interest to run static CFD cases with $\alpha = -5$ to compare how the comparative static stability derivatives would change with a central-difference calculation.

Figure 8 and 9 show the comparison of the SID models with the dynamic stability derivatives calculated from linear regression fits of the single oscillation dynamic maneuvers. In this case, some of the trends are captured, while others could have been captured better. However, even while some of these trends appear different, the apparent magnitude of the estimated dynamic derivatives appear correct, allowing for some intuition to be gained about the dynamic stability behavior of the aircraft. The strength of these plots is that the slopes predicted by the SID models can be compared between designs to show which design appears to be more/less stable.

The developed SID models are also used to simulate the integrated loads of some off-design maneuvers. These maneuvers are doublet-type maneuvers with very fast transitions between dwell periods, where conditions are held stable. The maneuvers excite transient dynamics that cannot be predicted by the quasi-steady models. However, the dwell portions of the doublets can be compared to the SID models to demonstrate the models’ accuracy. Two separate

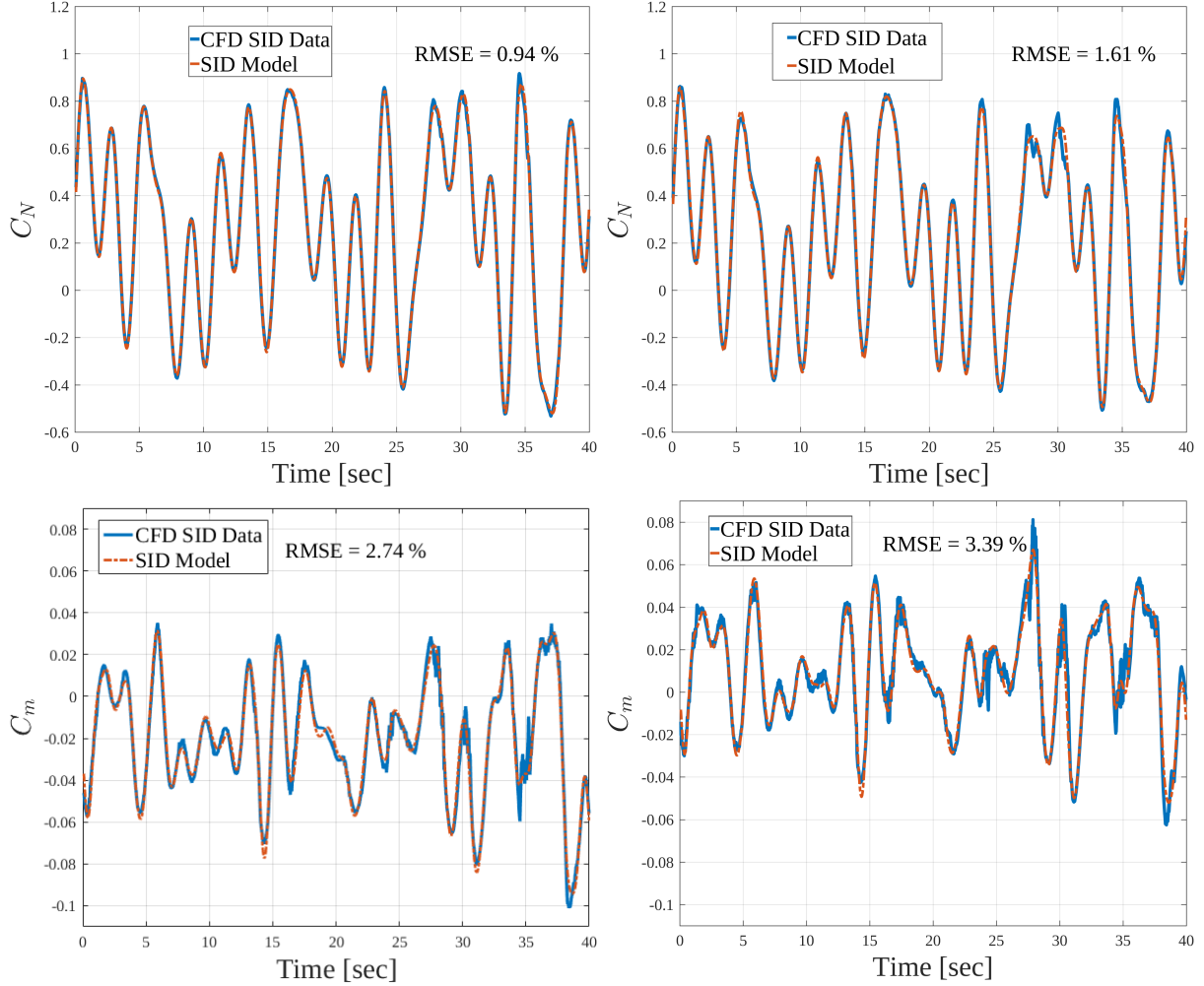


Fig. 4 C_N (upper) and C_m (lower) model fits of the 40 second training maneuver, with Muldicon design 1 (left) and design 2 (right).

doublet-type maneuvers are completed. One is a simple pitch doublet, and one is an IB doublet followed by an OB doublet. Both maneuvers are completed at Mach 0.5.

Figure 10 shows the C_N response to two doublet maneuvers. Only C_N results are shown. The SID predictions of the dwell portions of the doublet maneuvers quite close, with the SID models largely falling within a 10% band. The worst prediction of C_N is seen in the control surface doublet prediction for design 2. This can be indicative of either a ‘gap’ in the regressor coverage of the training maneuver, or possibly too many time-dependent effects in the training data caused by higher frequency motions.

IV. Summary

This work presents quasi-steady aerodynamic models that accurately and rapidly can generate aerodynamic data of a number of UCAVs. Schroeder-type forced motion were tested in this work. The motion covers the changes in angle of attack from -5° to 15° , a Mach range from 0.2 to 0.8, and inboard and outboard control surface deflections between $\pm 20^\circ$, while also independently modeling $\dot{\alpha}$ and pitch rate, q . The accuracy of the developed models were then compared to static data generated by CFD at test points across the flight-envelope, as well as against dynamic ‘off-design’ maneuvers performed in CFD. Good agreement was found in most cases, while the cost of simulating the forced motion is much less than the cost of generating all aerodynamic data for all points in the input space.

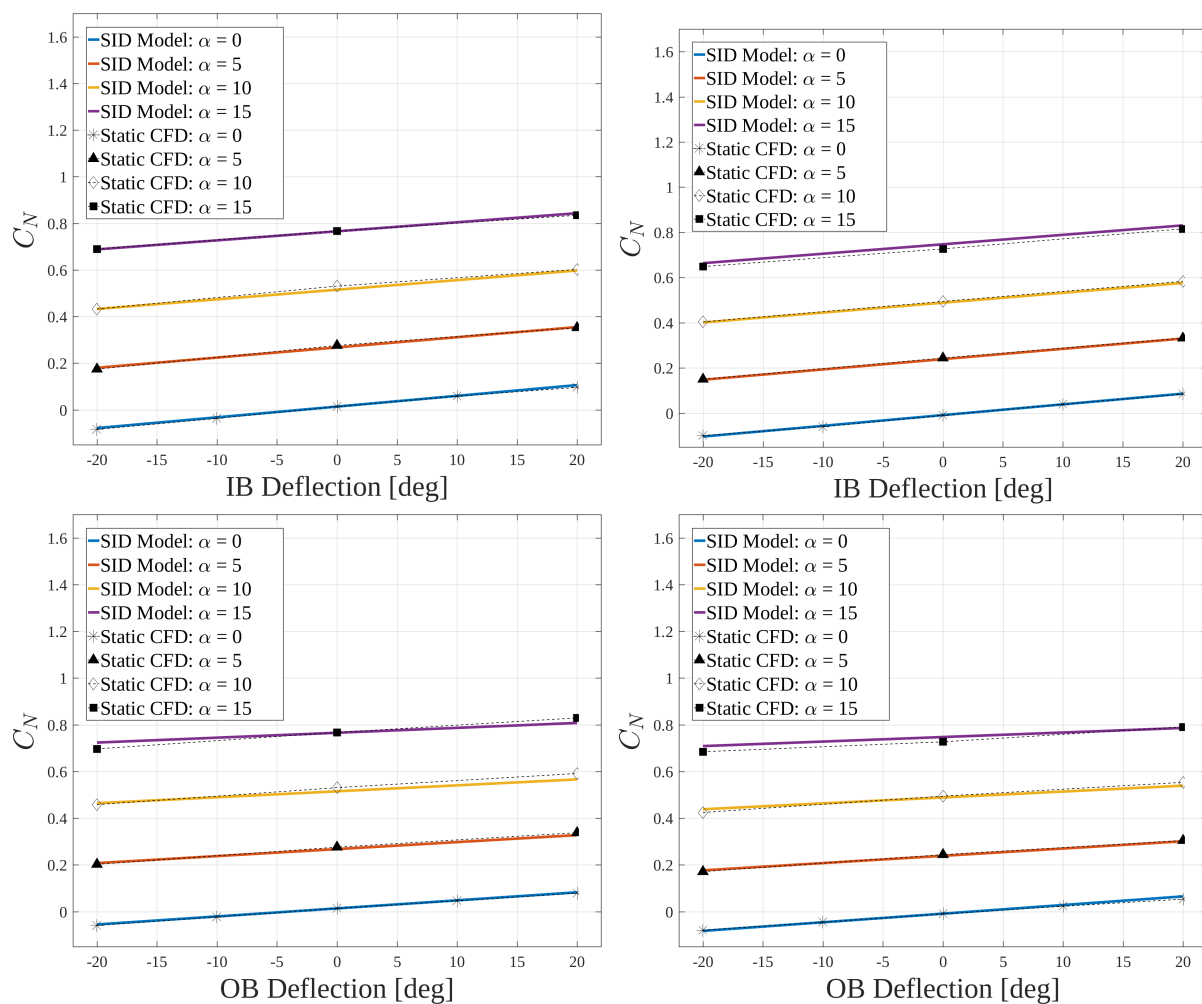


Fig. 5 C_N vs. IB deflections (upper) and C_N vs. OB deflections (lower) comparing static CFD and SID model for Muldicon design 1 (left) and design 2 (right) at varying α and Mach 0.4.

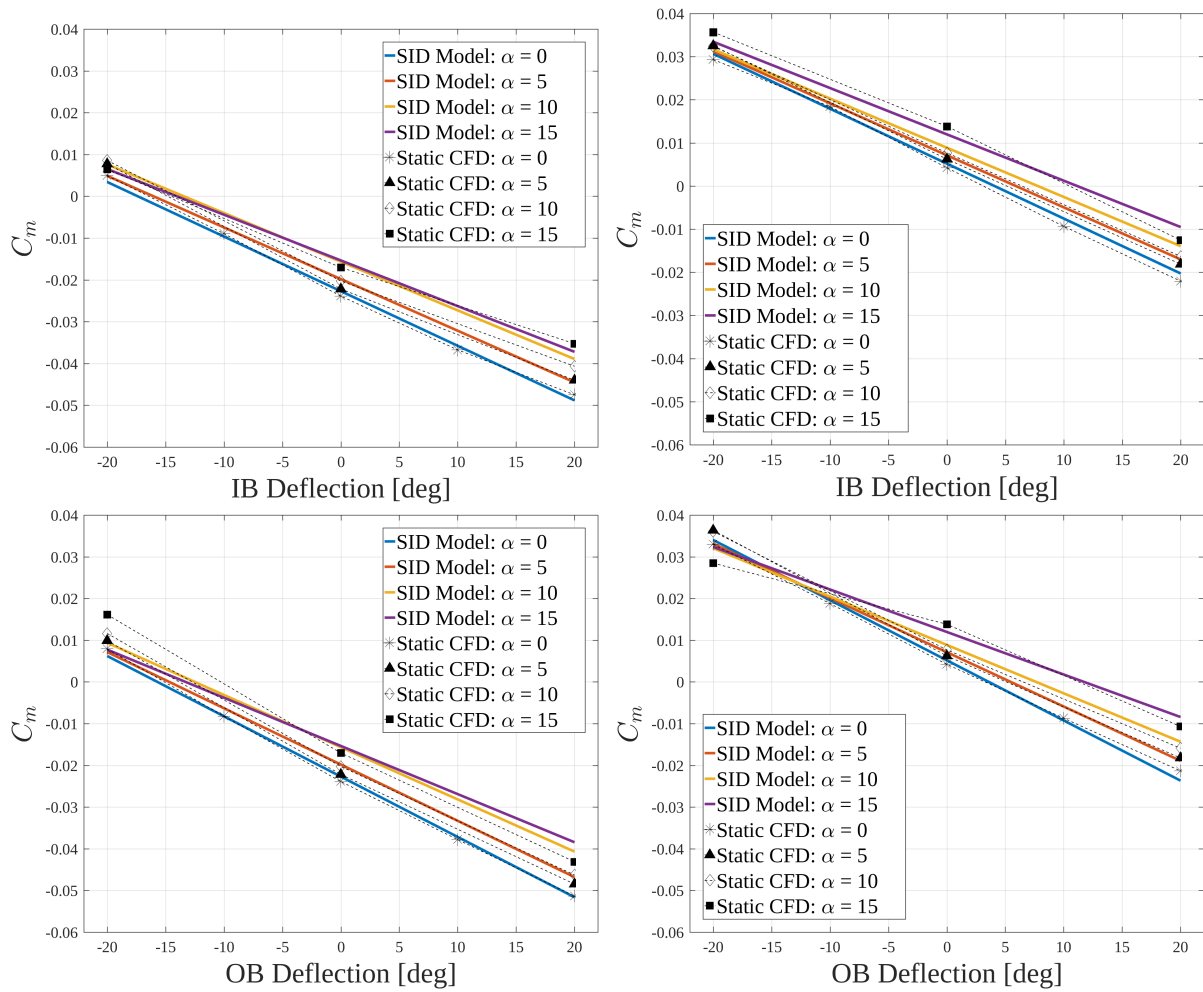


Fig. 6 C_m vs. IB deflections (upper) and C_m vs. OB deflections (lower) comparing static CFD and SID model for Muldicon design 1 (left) and design 2 (right) at varying α and Mach 0.4.

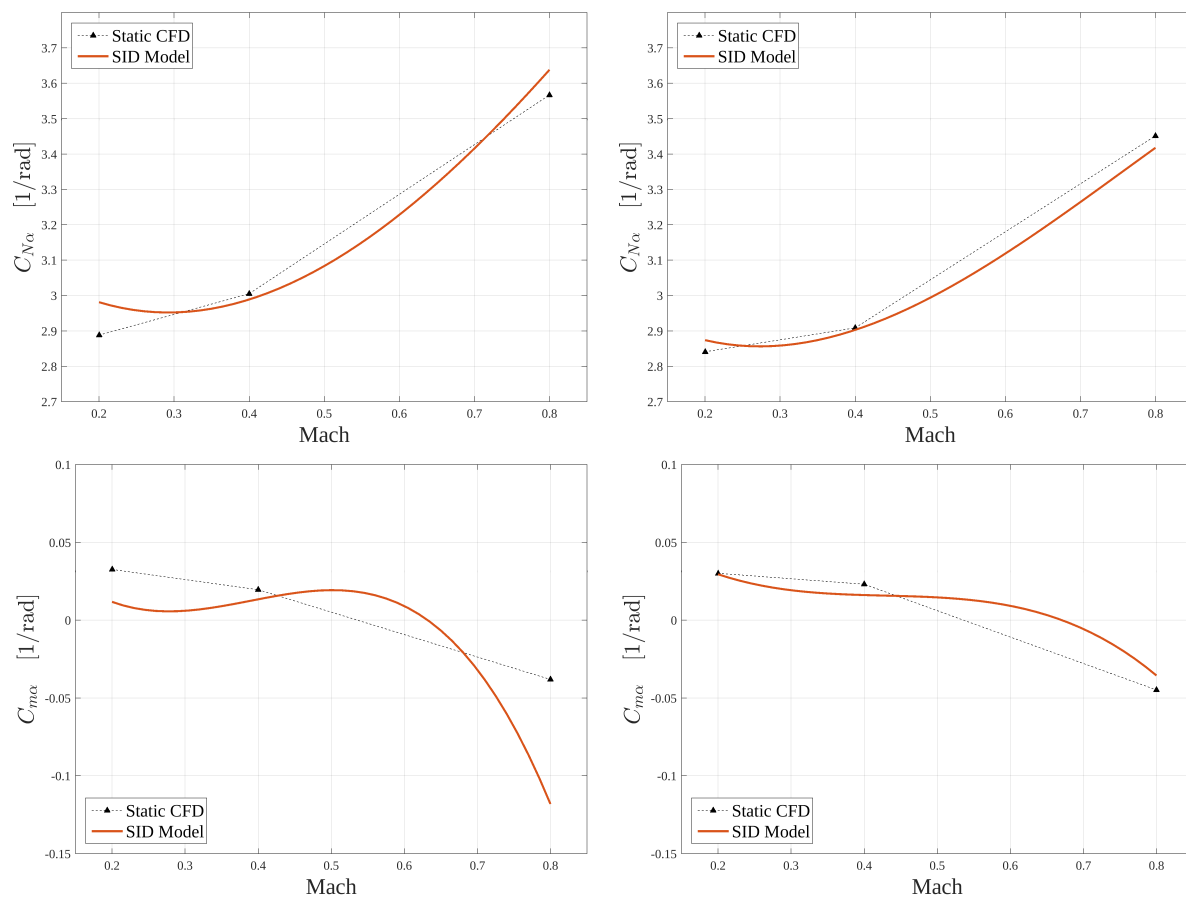


Fig. 7 $C_{N\alpha}$ (upper) and $C_{m\alpha}$ (lower) comparing static CFD and SID model for Design 1 (left) and Design 2 (right) over the tested Mach range.

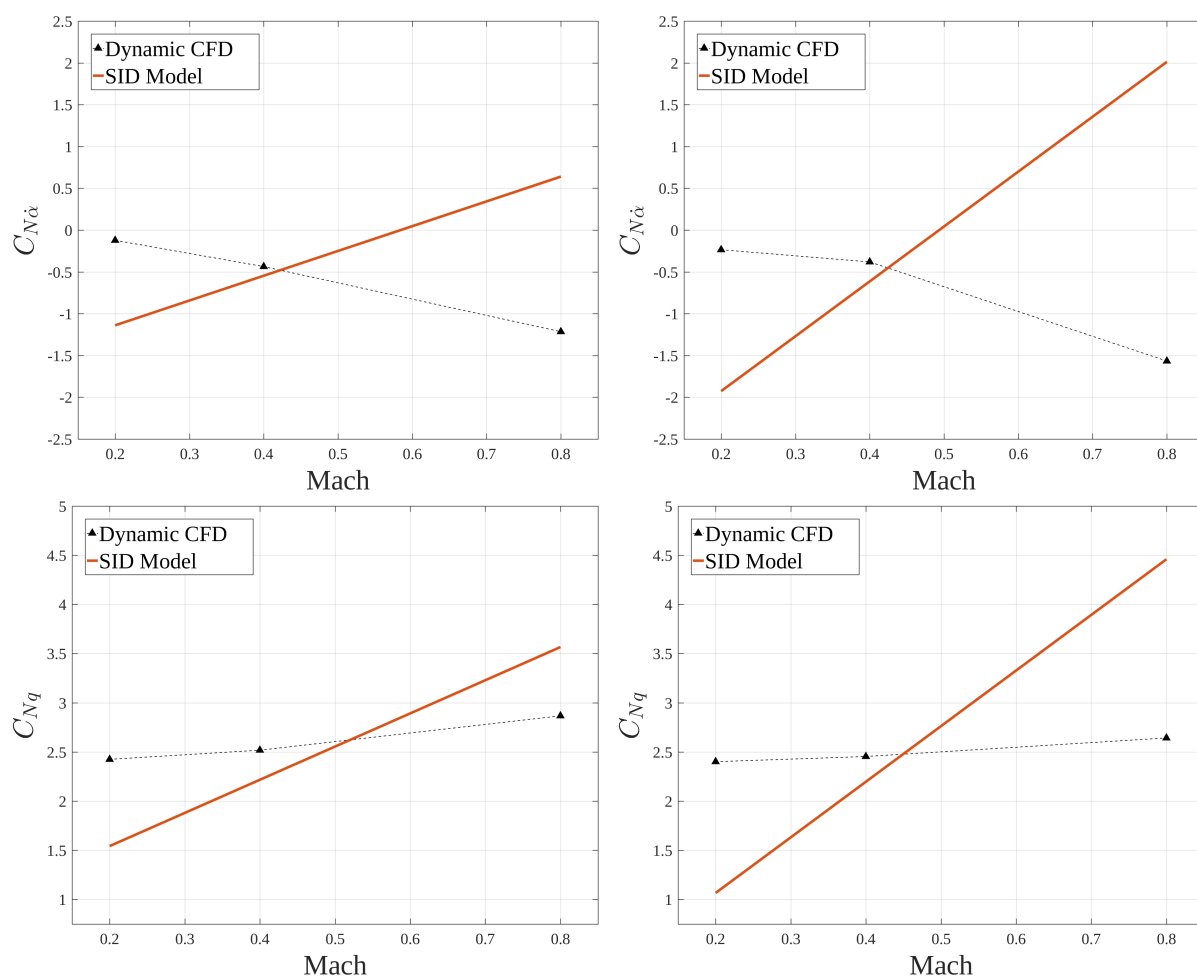


Fig. 8 $C_{N\dot{\alpha}}$ (upper) and C_{Nq} (lower) comparing static CFD and SID model for Design 1 (left) and Design 2 (right) over the tested Mach range.

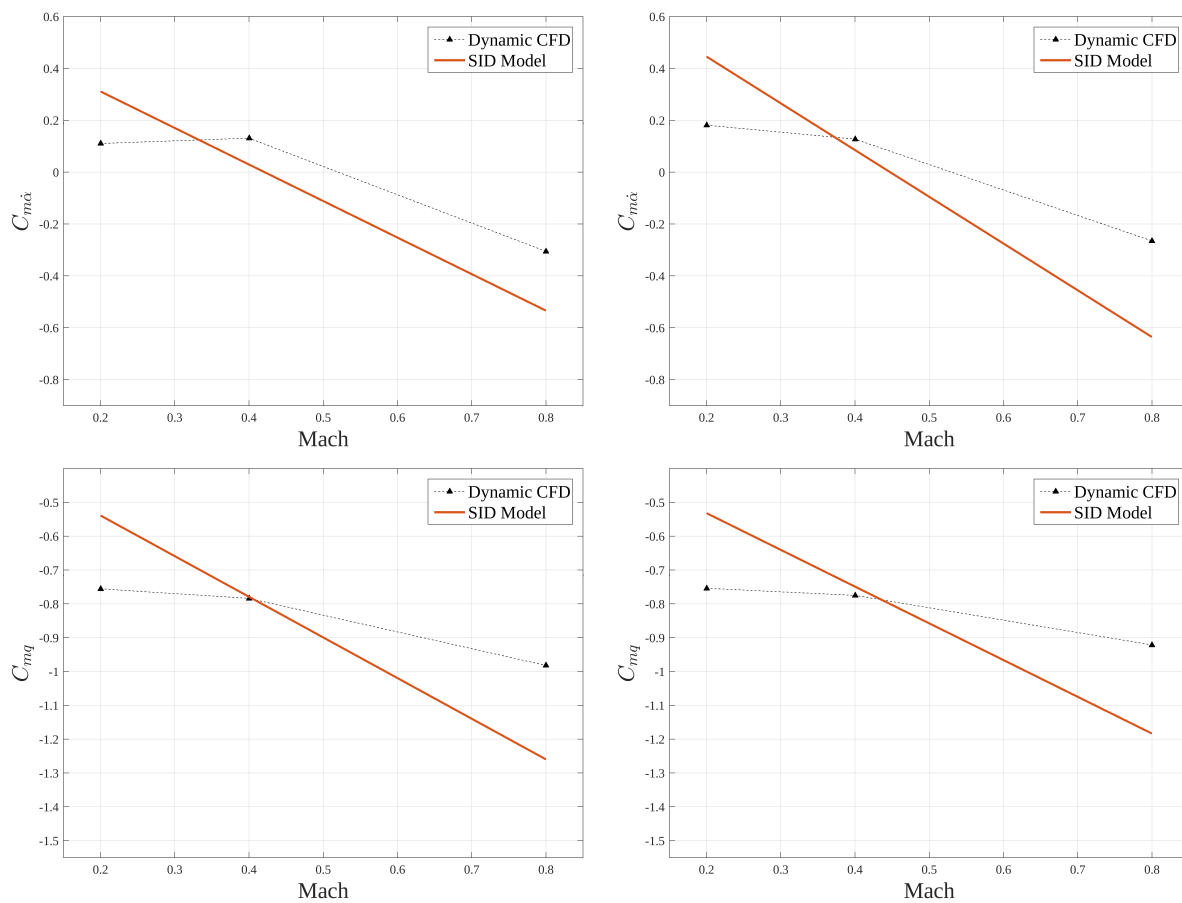


Fig. 9 $C_{m\alpha}$ (upper) and C_{mq} (lower) comparing static CFD and SID model for Design 1 (left) and Design 2 (right) over the tested Mach range.

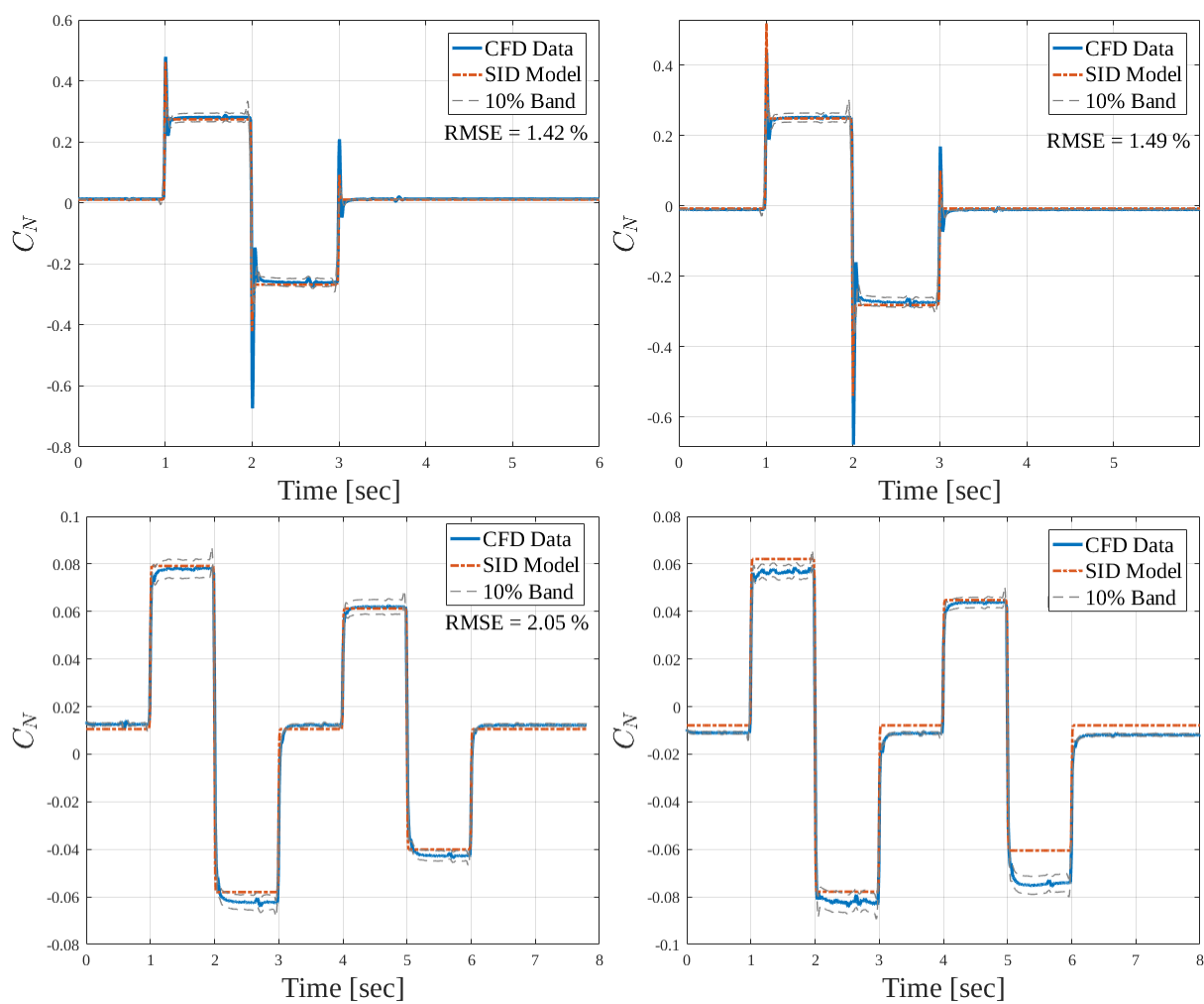


Fig. 10 The pitch doublet (upper) and the control surface doublet (lower) comparing off-design CFD and SID model for design 1 (left) and design 2 (right) at Mach 0.5.

The methods developed here allow for major differences between aircraft designs or configuration to be analyzed. The advantage of this method is that only a single training maneuver is required for each aircraft design to develop the model. Once developed, the models can predict everything from the basic integrated loads to the static and dynamic stability derivatives. This allows the major aerodynamic trends between candidate aircraft to be easily compared side by side.

V. Acknowledgements

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References

- [1] Ghoreyshi, M., and Cummings, R. M., "From Spreadsheets to Simulation-Based Aircraft Conceptual Design," *Applied Physics Research*, Vol. 6, No. 3, 2014, pp. Published by Canadian Center of Science and Education, 64–80.
- [2] Nelson, D. M., Irving, J. P., Ghoreyshi, M., Jirásek, A., and Lofthouse, A. J., "Experimental and Numerical Investigation of Flight Dynamics of a Generic Lambda Wing Configuration," *Aerospace Science and Technology*, Vol. 71, 2017, pp. 706–724.
- [3] Ghoreyshi, M., Badcock, K. J., and Woodgate, M. A., "Accelerating the Numerical Generation of Aerodynamic Models for Flight Simulation," *Journal of Aircraft*, Vol. 46, No. 3, 2009, pp. 972–980.
- [4] Ghoreyshi, M., Lofthouse, A. J., Findaly, D. B., and Lee, J., "Indicial Methods for the Numerical Calculation of Dynamic Derivatives," *AIAA Journal*, Vol. 55, No. 7, 2017, pp. 2278 – 2294.
- [5] Allen, J., and Ghoreyshi, M., "Forced Motion Design for Aerodynamic Identification and Modeling of a Generic Missile Configuration," *Aerospace Science and Technology*, Vol. 77, 2018, pp. 742 – 754.
- [6] Allen, J. D., *Aerodynamic Identification and Modeling of Generic UCAV Configurations with Control Surface Integration*, M.S. thesis, University of Colorado Colorado Springs, 2017.
- [7] Cummings, R. M., Liersch, C. M., and Schütte, A., "Multi-Disciplinary Design and Performance Assessment of Effective, Agile NATO Air Vehicles," *Submitted to AIAA Aviation and Aeronautics Forum and Exposition, Atlanta, Georgia*, 2018.
- [8] Nangia, R. K., Coppin, J., and Ghoreyshi, M., "UCAV Wing Design, Assessment and Comparisons," *Submitted to AIAA Aviation and Aeronautics Forum and Exposition, Atlanta, Georgia*, 2018.
- [9] Greenwell, D. I., "Frequency Effects on Dynamic Stability Derivatives Obtained from Small-Amplitude Oscillatory Testing," *Journal of Aircraft*, Vol. 35, No. 5, 1998, pp. 776–783.
- [10] Klein, V., and Morelli, E. A., *Aircraft System Identification - Theory and Practice*, American Institute of Aeronautics and Astronautics, Reston, VA, 2006.
- [11] Computational Research and Engineering Acquisition Tools And Environments (CREATE) - Air Vehicles, *HPCMP CREATEKestrel User's Guide, Version 7.0*, Eglin AFB, FL, Sept 2016.
- [12] Schroeder, M. R., "Synthesis of Low-Peak-Factor Signals and Binary Sequences With Low Autocorrelation," *IEEE Transactions on Information Theory*, January, 1970.
- [13] Morelli, E. A., "Flight Test Maneuvers for Efficient Aerodynamic Modeling," *American Institute of Aeronautics and Astronautics*, Vol. 49, 2012.
- [14] Morelli, E. A., "Efficient Global Aerodynamic Modeling from Flight Data," *50th AIAA Aerospace Sciences Meeting, Nashville, TN*, 2012.
- [15] Morelli, E. A., "System IDentification Programs for AirCRAFT (SIDPAC)," *AIAA Paper 2002-4704*, August, 2002.

- [16] Rein, M., Irving, J. P., Rigby, G., and Birch, T. J., "High speed static experimental investigations to estimate control device effectiveness and S&C capabilities," *32nd AIAA Applied Aerodynamics Conference*, 16-20 June 2014.
- [17] Morton, S. A., and McDaniel, D. R., "CFD-Based Model Building of the F-16XL Static and Dynamic Loads Using Kestrel," *55th AIAA Aerospace Sciences Meeting*, AIAA 2017-0286.

Appendix

Some flow visualization figures of Muldicon Design 3 wing and predictions of SID models are shown in Figures 11- 15.

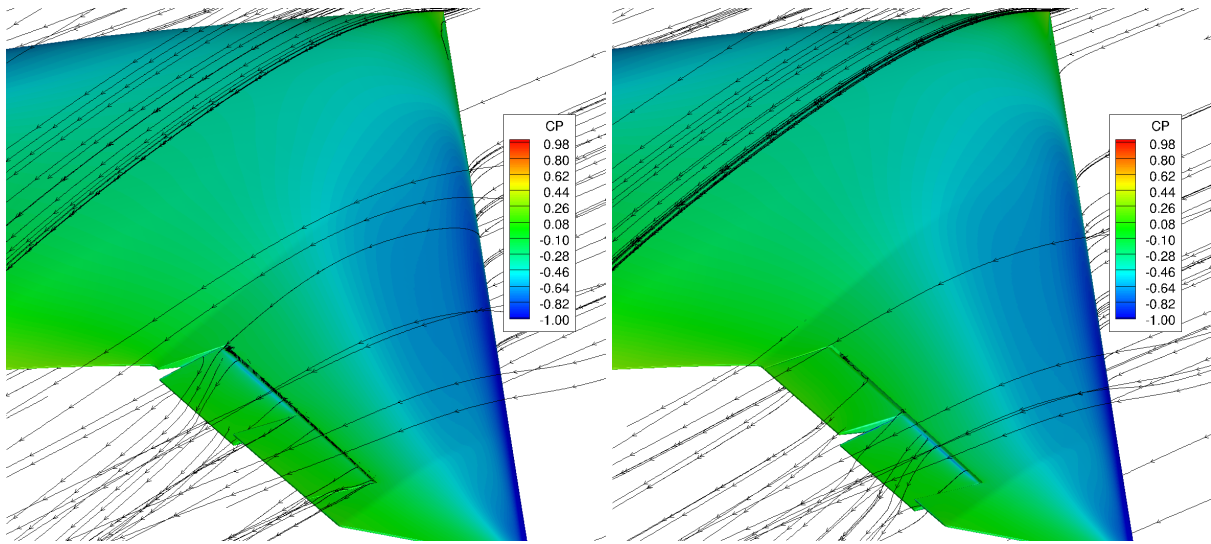


Fig. 11 Design 3 surface solutions for ten degrees inboard flap (left) and outboard flap (right) flap deflections at $\alpha = 8^\circ$ and Mach 0.2.

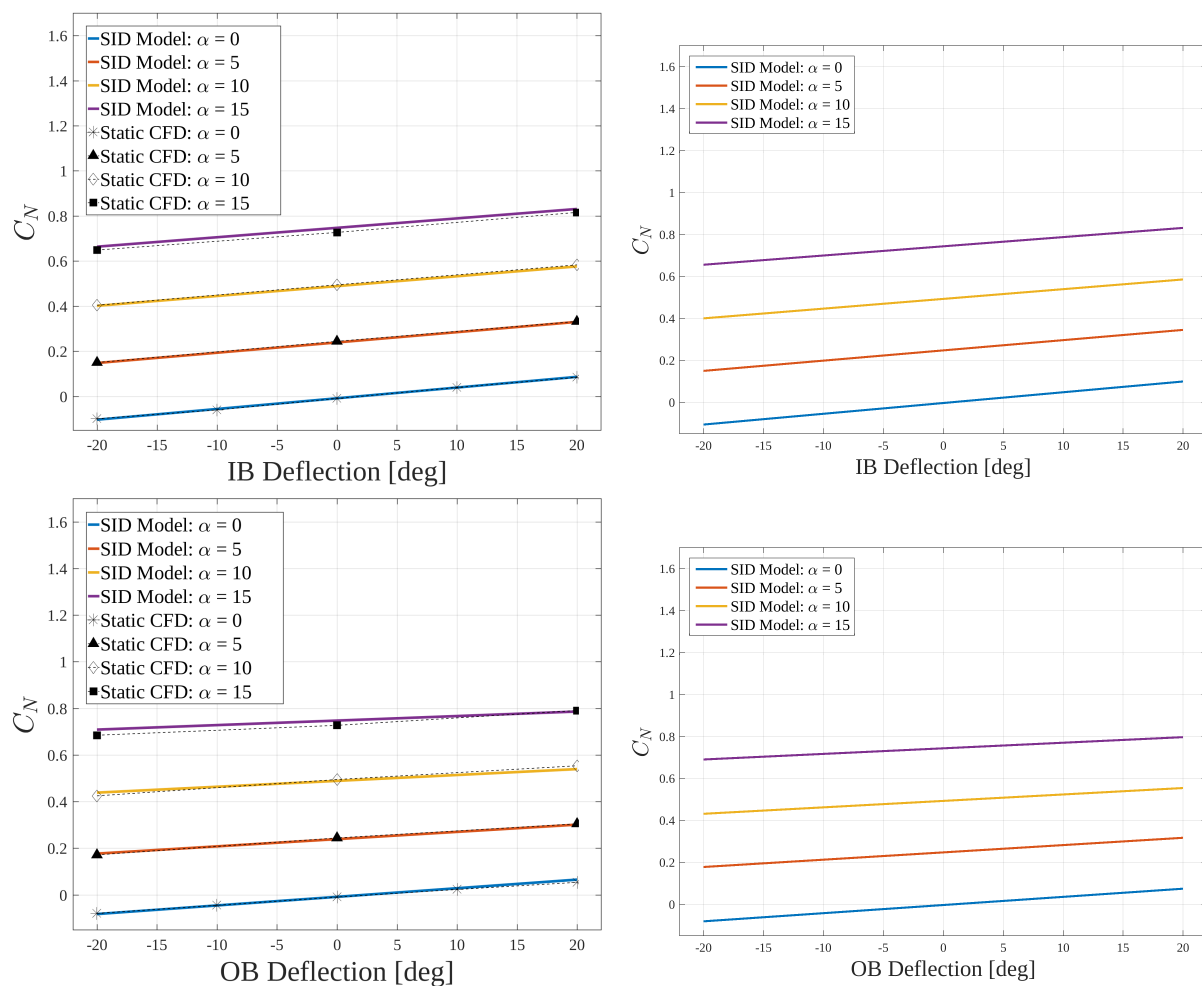


Fig. 12 C_N vs. IB deflections (upper) and C_N vs. OB deflections (lower) comparing SID models for Muldicon design 2 (left) and design 3 (right) at varying α and Mach 0.4.

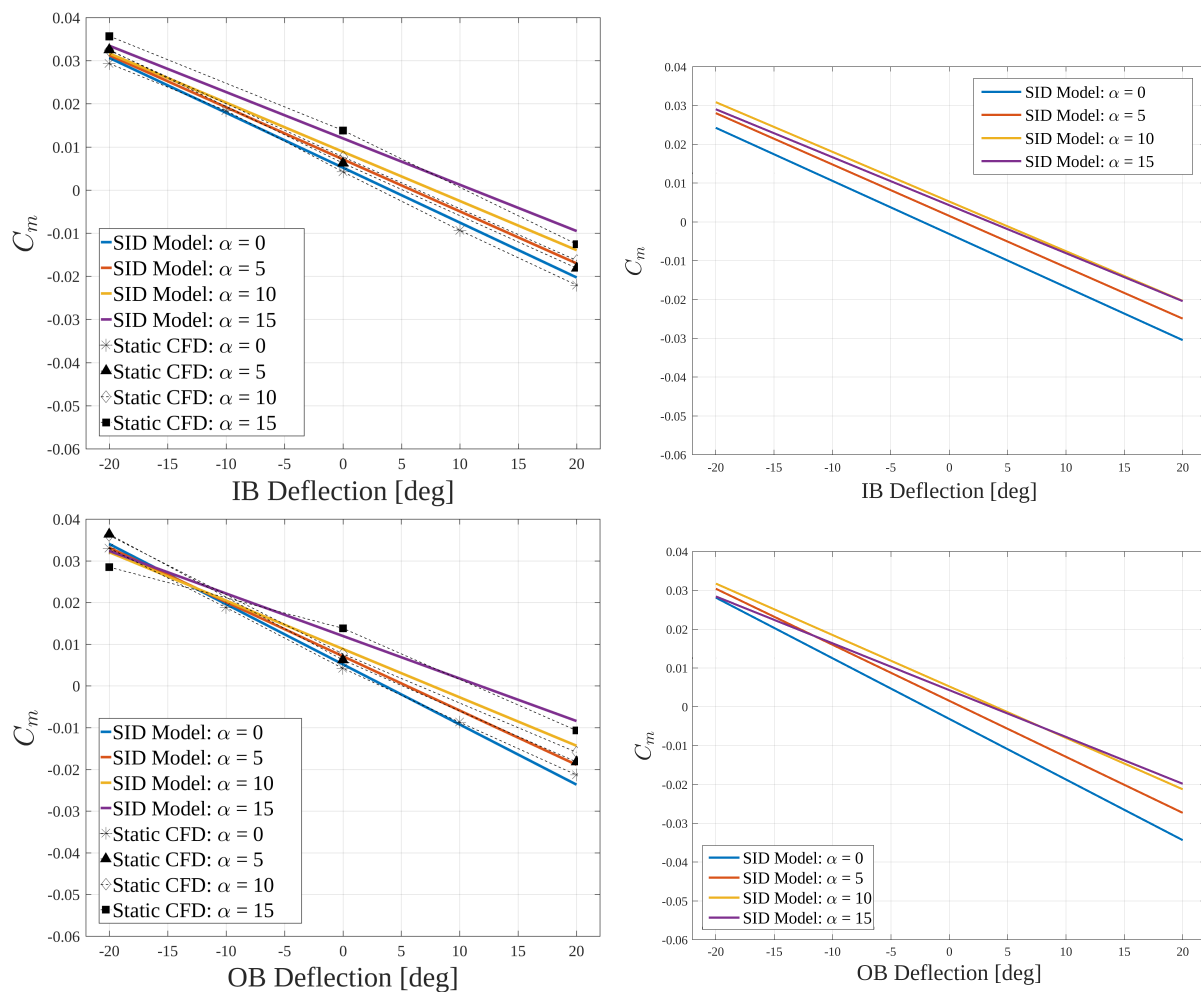


Fig. 13 C_m vs. IB deflections (upper) and C_m vs. OB deflections (lower) comparing SID models for Muldicon design 2 (left) and design 3 (right) at varying α and Mach 0.4.

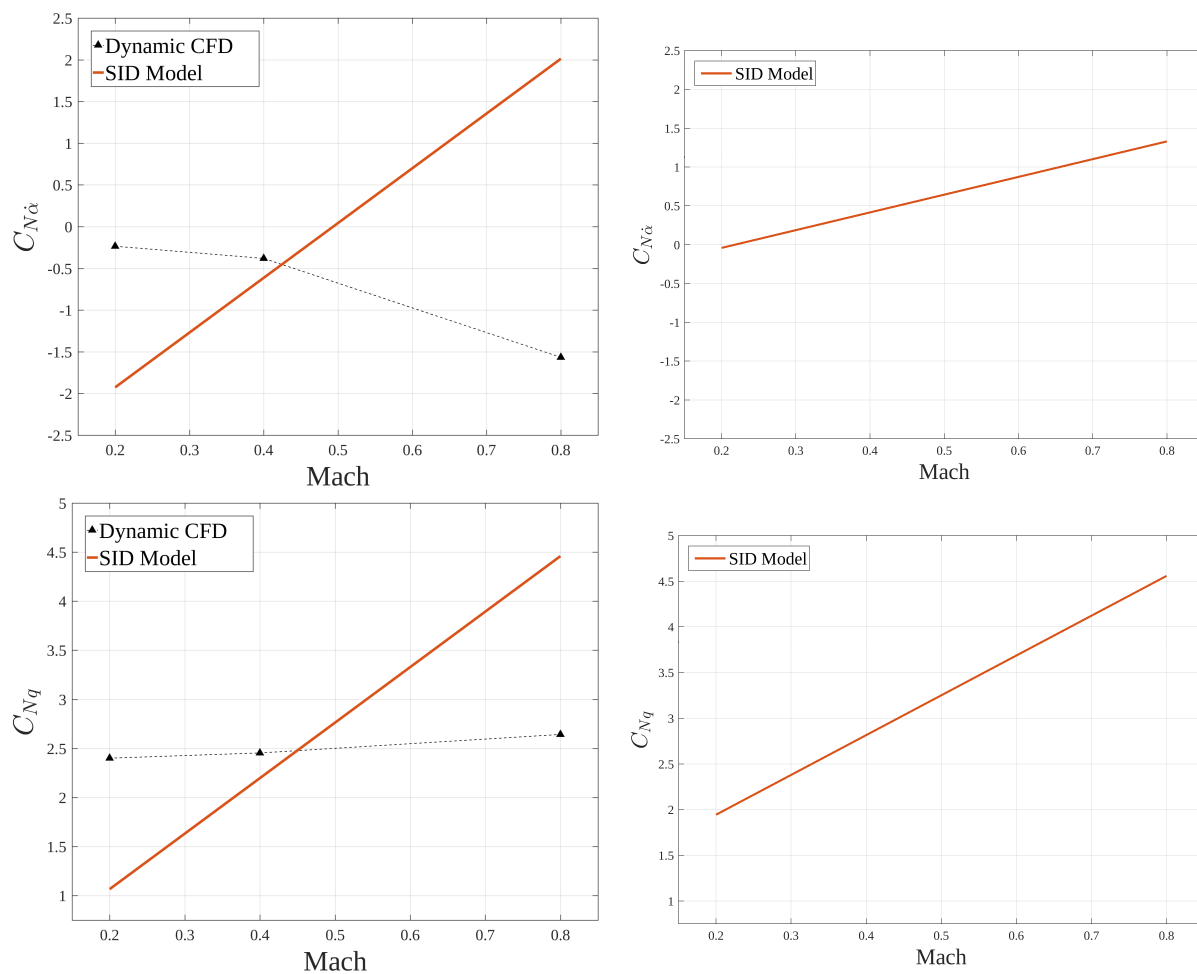


Fig. 14 $C_{N\dot{\alpha}}$ (upper) and C_{Nq} (lower) comparing static SID models for Muldicon design 2 (left) and design 3 (right) over the tested Mach range.

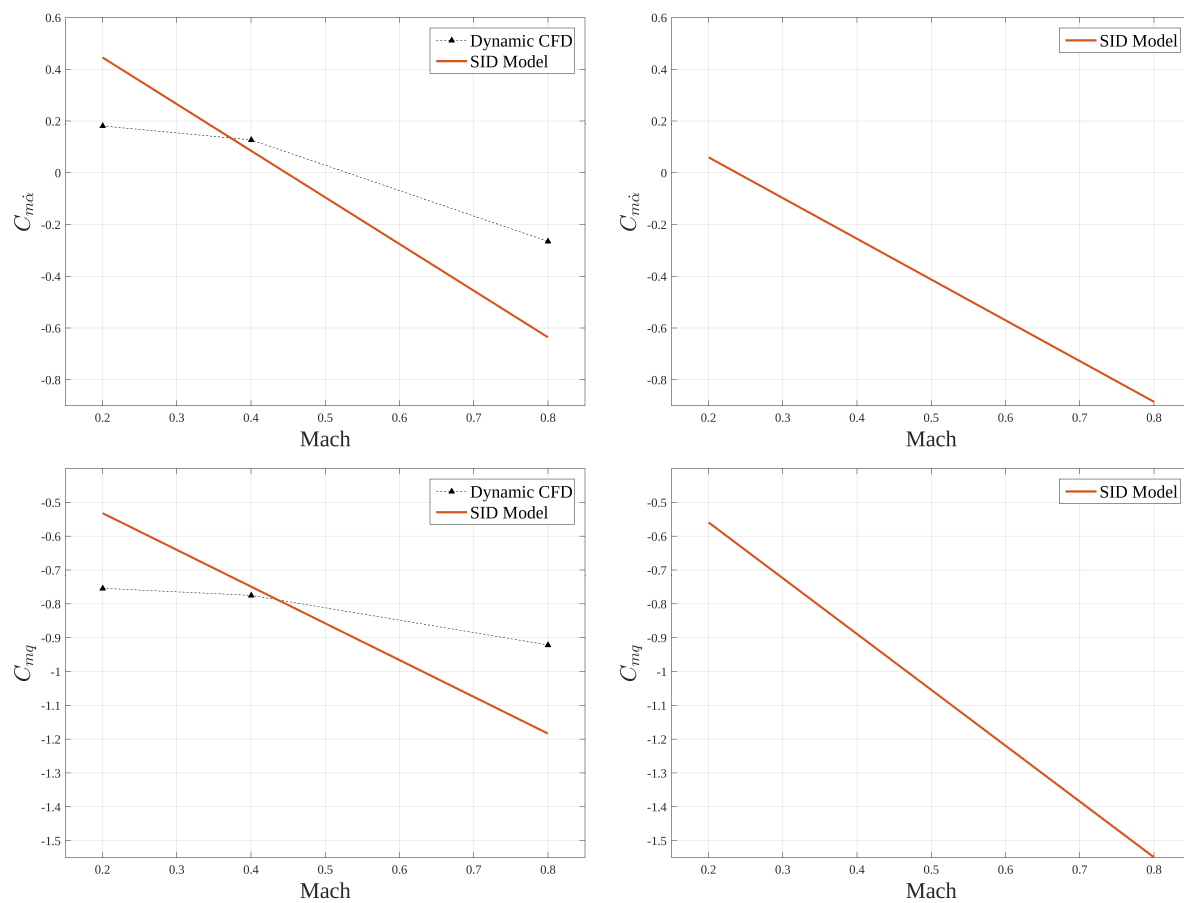


Fig. 15 $C_{m\dot{\alpha}}$ (upper) and C_{mq} (lower) comparing SID models for design 2 (left) and design 3 (right) over the tested Mach range.